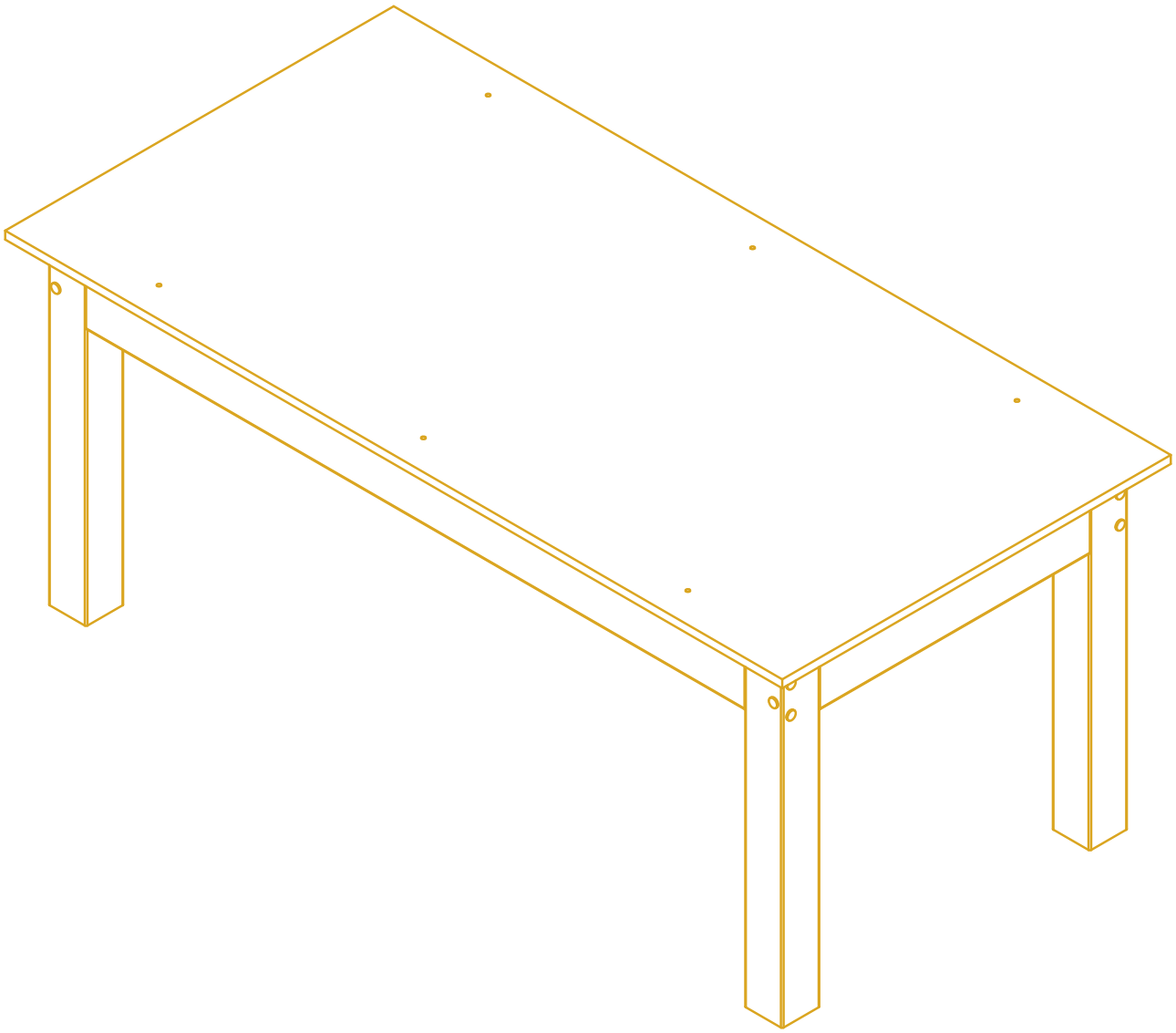
imperial-desk-1



A desk: a 3/4 in. plywood top on four 4x4 legs, braced by a 2x6 apron on each side and screwed together. Everything is cut from 8 ft. lumber and a 4x8 sheet, so the top is at most 48 x 96 in. and the desk at most 96 in. high.

Package: //pub/furniture/workspace/basic
Assemblies: 2
Steps: 8
Size: 1829 x 914 x 762 mm (72 x 36 x 30 in.), along X, Y and Z

Generated by PartCAD

Bill of Materials

Sub-Assemblies

//pub/furniture/workspace/basic

Assembly	Count	Description
imperial-desk-1/side; apron=25.0,leg=29.28125	2	An end of the desk: two legs joined by a side apron, flush with their tops and their outside faces and screwed through the legs. The desk takes two.

Parts

//pub/furniture/workspace/basic

Part	Count	Description
leg;length=29.28125	4	Leg, 4x4 cut to length
long-apron;length=61.0	2	Long apron, 2x6 cut to length
side-apron;length=25.0	2	Side apron, 2x6 cut to length
top;length=72,width=36	1	Top, 3/4 in. plywood cut to size

//pub/svc/commerce/homedepot

Part	Count	Description
screws/sdws22600db	16	Simpson Strong-Tie 0.220 in. x 6 in. T40 Washer Head Strong-Drive SDWS Timber Screw, Exterior Grade (12-Pack)
screws/spax-8x2	6	SPAX #8 x 2 in. T-Star Plus Drive Flat-Head Partial Thread Yellow Zinc Coated Multi-Material Screw (161-Box)

Stock

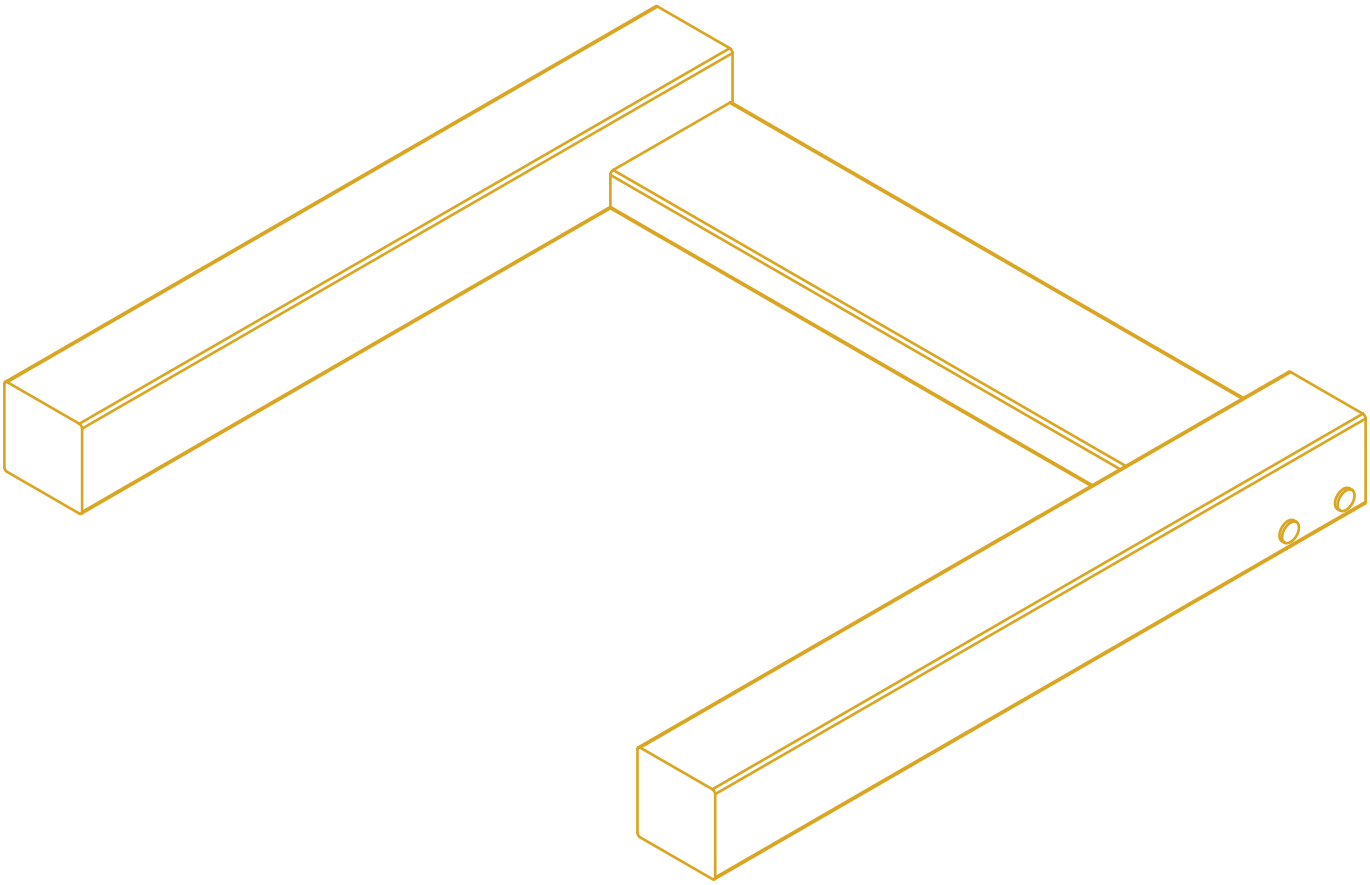
What the manufactured parts are made from: one piece for each part made from it.

//pub/std/imperial/dimensional-lumber/stock

Stock	Count	For	Description
lumber/2x6x8	4	side-apron;length=25.0, long-apron;length=61.0	2x6, 8 ft. (1-1/2 x 5-1/2 x 96 in.)
lumber/4x4x8	4	leg;length=29.28125	4x4, 8 ft. (3-1/2 x 3-1/2 x 96 in.)
plywood/3-4x4x8	1	top;length=72,width=36	3/4 in. plywood, 4 x 8 ft. (sold as 23/32 in.; 23/32 x 48 x 96 in.)

SUB-ASSEMBLY - MAKE 2

imperial-desk-1/side



An end of the desk: two legs joined by a side apron, flush with their tops and their outside faces and screwed through the legs. The desk takes two.

Package: //pub/furniture/workspace/basic
Size: apron 25, leg 29.28125
Steps: 2
Needed: 2

The build needs 2 of these. Repeat this section 2 times - the steps are the same every time.

What goes into one

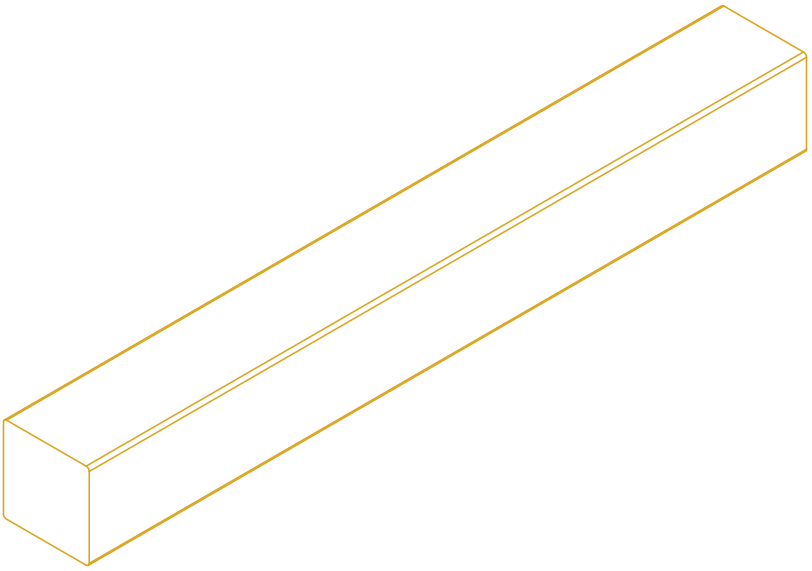
Item	Count	Description
leg	2	Leg, 4x4 cut to length (length 29.28125)
side-apron	1	Side apron, 2x6 cut to length (length 25)
//pub/svc/commerce/homedepot:screws/sdws22600db	4	Simpson Strong-Tie 0.220 in. x 6 in. T40 Washer Head Strong-Drive SDWS Timber Screw, Exterior Grade (12-Pack)

Start with leg-1.

Lay a leg on the floor. The face on the floor will be the outside of the desk.

MAKE 4

leg



Leg, 4x4 cut to length

Package: //pub/furniture/workspace/basic
Needed: 4
Size: length 29.28125
Made from: //pub/std/imperial/dimensional-lumber/stock:lumber/4x4x8

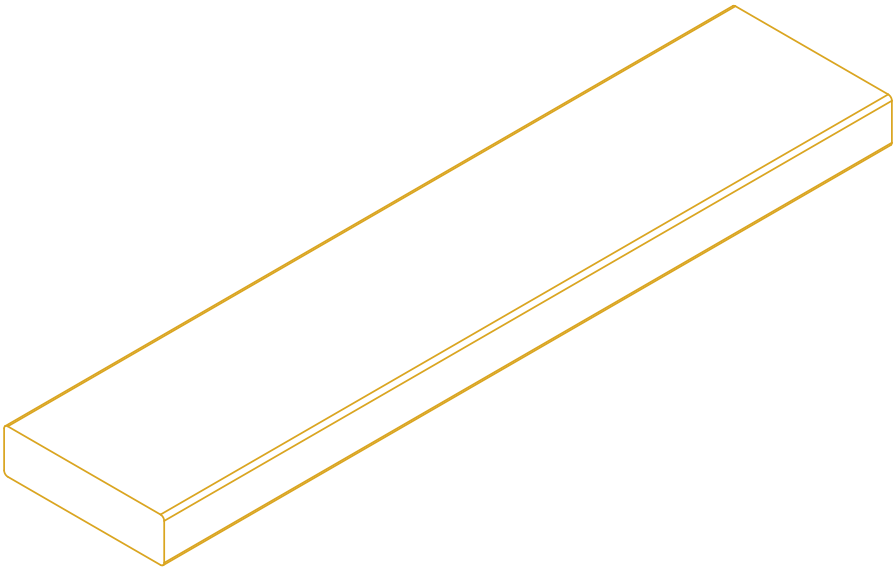
Made by taking material away from `//pub/std/imperial/dimensional-lumber/stock:lumber/4x4x8`.

Cut it to size with a saw:

1. Cut across Y, 743.744 mm (29.281 in) from the -Y end of the stock.

MAKE 2

side-apron



Side apron, 2x6 cut to length

Package: //pub/furniture/workspace/basic
Needed: 2
Size: length 25
Made from: //pub/std/imperial/dimensional-lumber/stock:lumber/2x6x8

Made by taking material away from `//pub/std/imperial/dimensional-lumber/stock:lumber/2x6x8`.

Cut it to size with a saw:

1. Cut across Y, 635 mm (25 in) from the -Y end of the stock.

Step 1 of 2

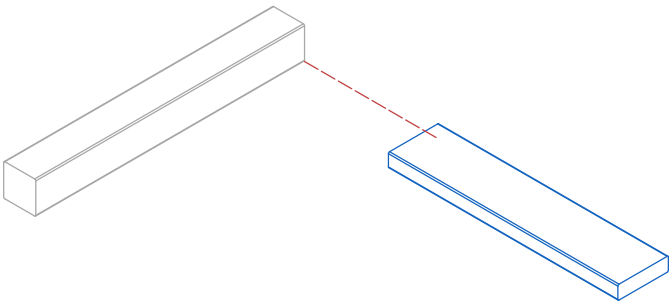
Parts for this step



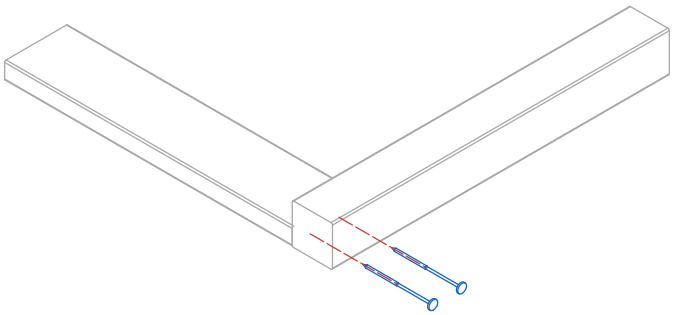
2 × screws/sdws22600db

Lay the side apron flat on the floor and butt its end against the side of the leg, flush with the top end of the leg.

Drive two screws through the leg into the end of the apron, 1 in. and 3-1/2 in. from the top end of the leg and 3/4 in. up from the floor.



1. Put it in place



2. Fasten it

Joint: apron end-a to leg-1 rail-a, pushed into place, up to 20 N.

Fasteners: 2 × screws/sdws22600db into apron through leg-1, driven in, cutting its own thread.

Step 2 of 2

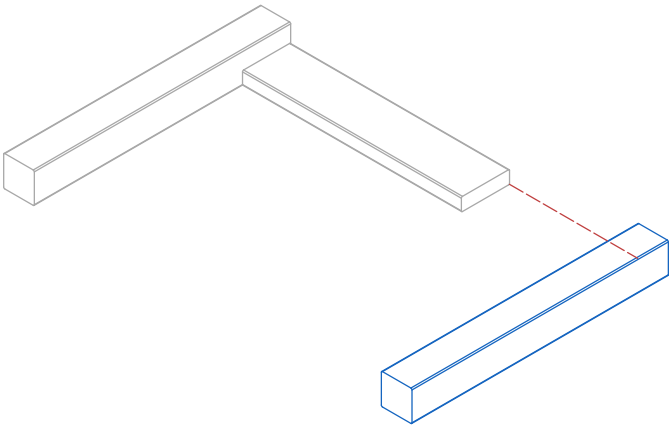
Parts for this step



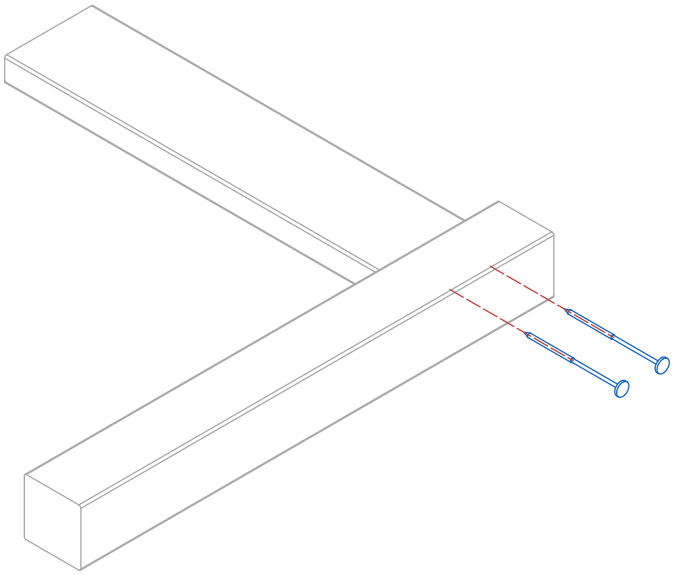
2 × screws/sdws22600db

Butt the second leg against the other end of the apron the same way, its top end flush with the first leg's.

Drive two screws through this leg into the apron the same way.



1. Put it in place

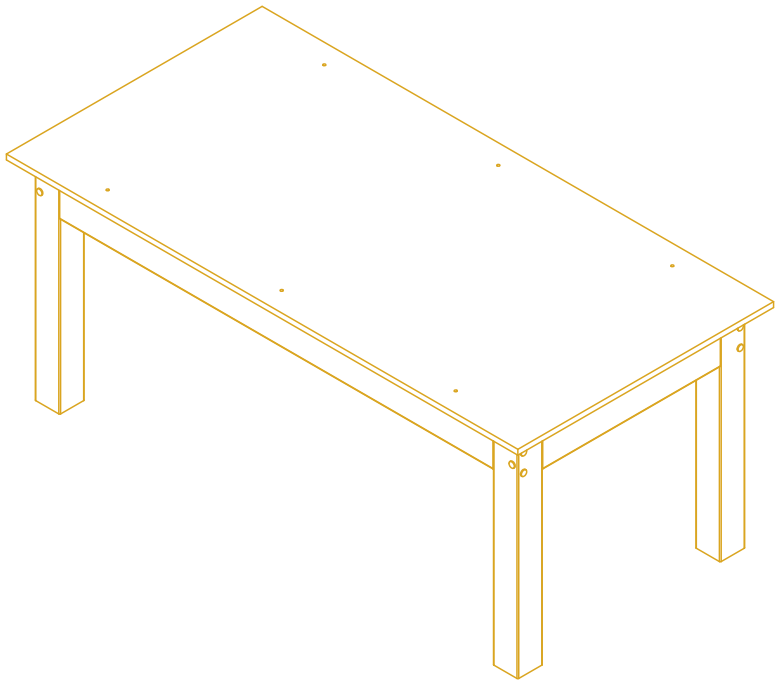


2. Fasten it

Joint: leg-2 rail-b to apron end-b, pushed into place, up to 20 N.

Fasteners: 2 × screws/sdws22600db into apron through leg-2, driven in, cutting its own thread.

imperial-desk-1



A desk: a 3/4 in. plywood top on four 4x4 legs, braced by a 2x6 apron on each side and screwed together. Everything is cut from 8 ft. lumber and a 4x8 sheet, so the top is at most 48 x 96 in. and the desk at most 96 in. high.

Package: //pub/furniture/workspace/basic
Steps: 4

What goes into it

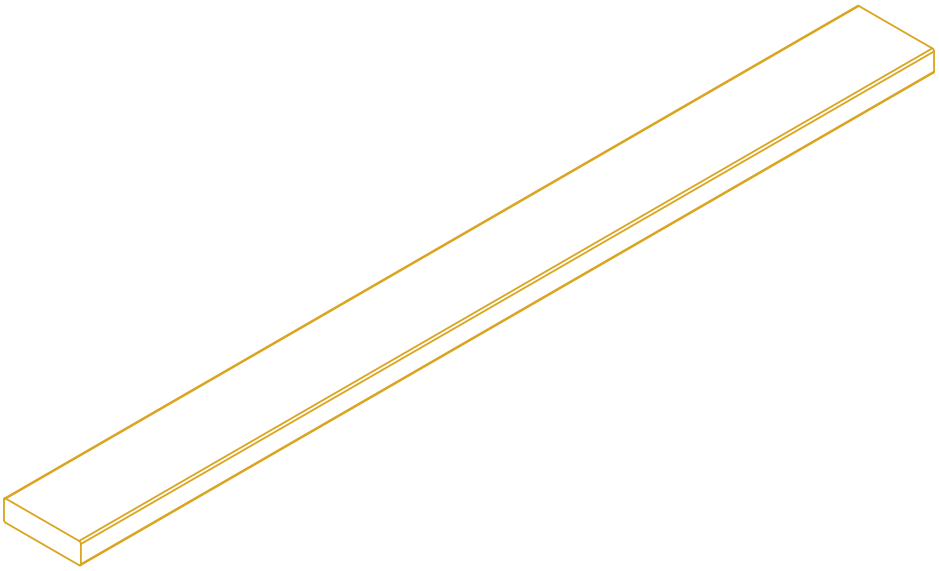
Item	Count	Description
imperial-desk-1/side	2	An end of the desk: two legs joined by a side apron, flush with their tops and their outside faces and screwed through the legs. The desk takes two. (apron 25, leg 29.28125)
long-apron	2	Long apron, 2x6 cut to length (length 61)
//pub/svc/commerce/homedepot:screws/sdws22600db	8	Simpson Strong-Tie 0.220 in. x 6 in. T40 Washer Head Strong-Drive SDWS Timber Screw, Exterior Grade (12-Pack)
top	1	Top, 3/4 in. plywood cut to size (length 72, width 36)
//pub/svc/commerce/homedepot:screws/spax-8x2	6	SPAX #8 x 2 in. T-Star Plus Drive Flat-Head Partial Thread Yellow Zinc Coated Multi-Material Screw (161-Box)

Start with left-side.

Stand an end frame on its feet, with its outside face (the one the apron is flush with) to the left. Have a helper hold it.

MAKE 2

long-apron



Long apron, 2x6 cut to length

Package: //pub/furniture/workspace/basic
Needed: 2
Size: length 61
Made from: //pub/std/imperial/dimensional-lumber/stock:lumber/2x6x8

Made by taking material away from `//pub/std/imperial/dimensional-lumber/stock:lumber/2x6x8`.

Cut it to size with a saw:

1. Cut across Y, 1549.4 mm (61 in) from the -Y end of the stock.

Step 1 of 4

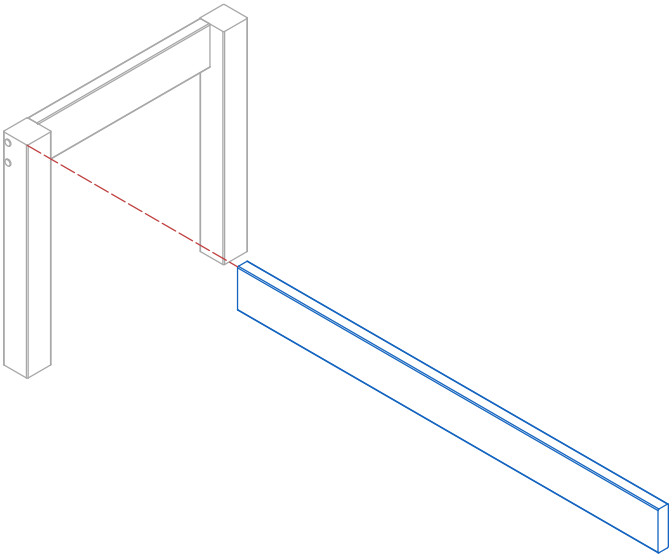
Parts for this step



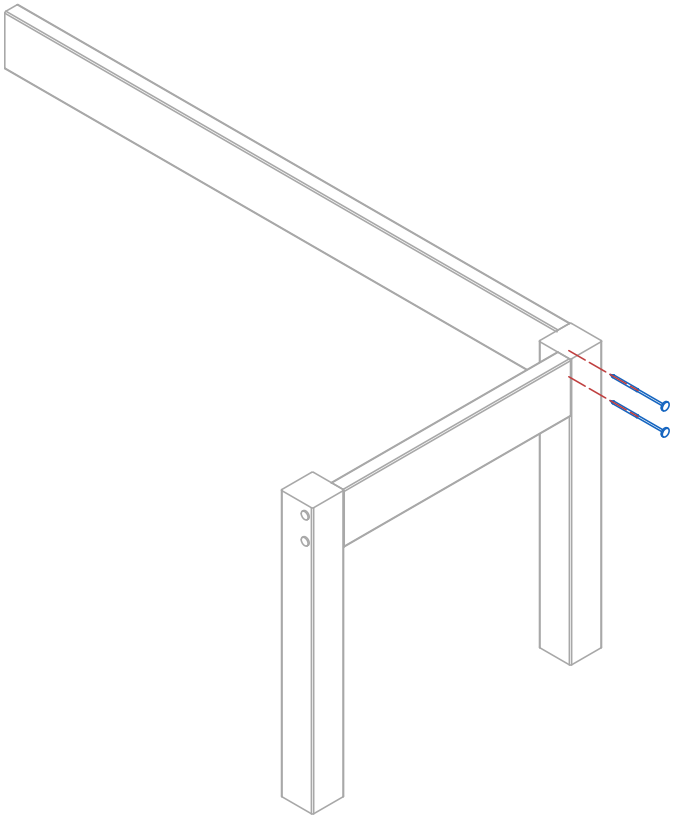
2 × screws/sdws22600db

Butt a long apron against the inside face of the front leg, flush with the front of the leg and with its top.

Drive two screws through the leg, from the outside of the desk, into the end of the apron, 2 in. and 4-1/2 in. below the top and 3/4 in. in from the front.



1. Put it in place



2. Fasten it

Joint: front-apron end-b to left-side rail-1, pushed into place, up to 20 N.

Fasteners: 2 × screws/sdws22600db into front-apron through left-side, driven in, cutting its own thread.

Step 2 of 4

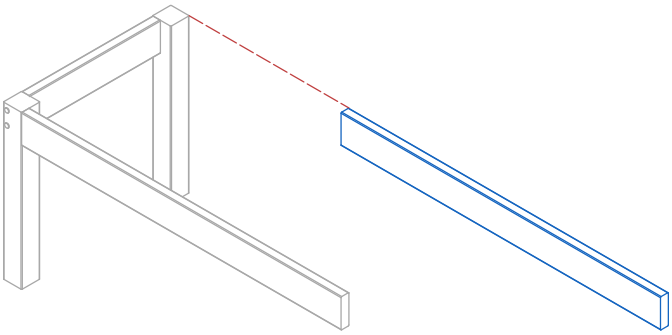
Parts for this step



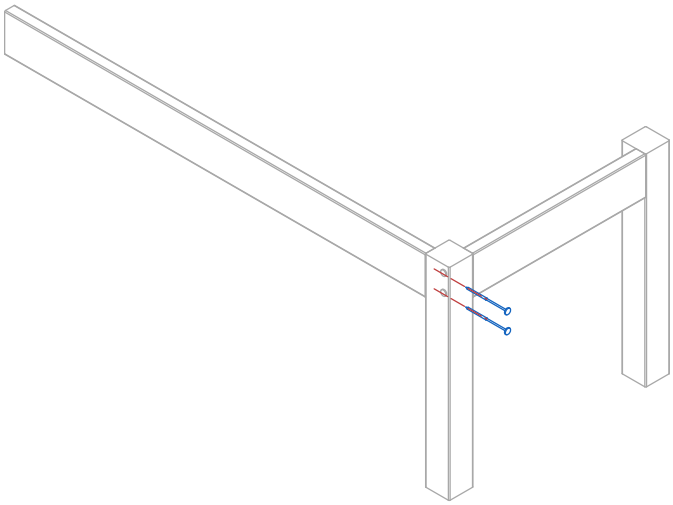
2 × screws/sdws22600db

Butt the other long apron against the back leg the same way, flush with the back of the leg and with its top.

Screw it on the same way.



1. Put it in place



2. Fasten it

Joint: back-apron end-a to left-side rail-2, pushed into place, up to 20 N.

Fasteners: 2 × screws/sdws22600db into back-apron through left-side, driven in, cutting its own thread.

Step 3 of 4

Parts for this step

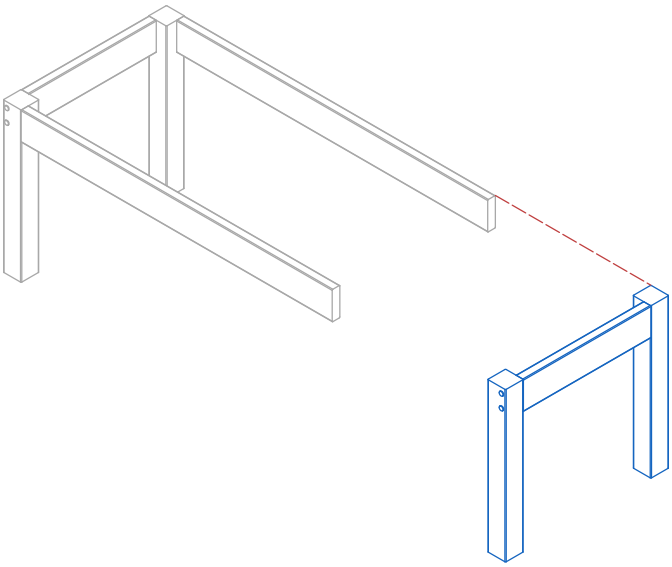


4 × screws/sdws22600db

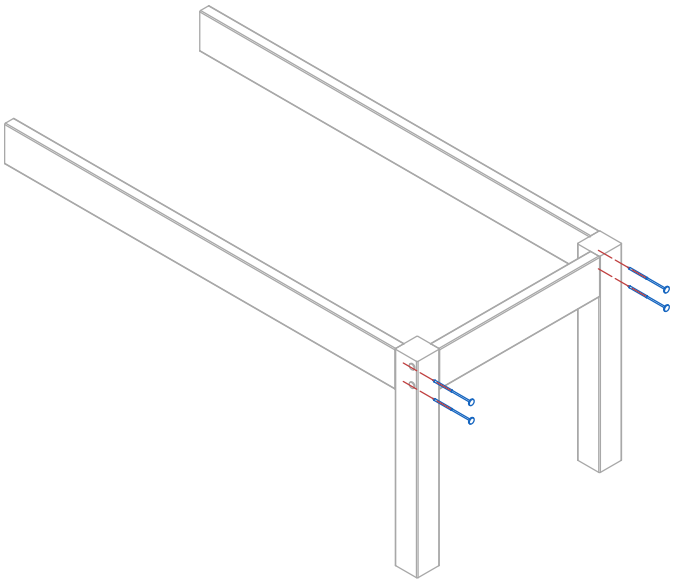
Stand the other end frame against the free ends of the long aprons, with its outside face to the right, flush with the aprons' tops.

It also meets front-apron.

Screw both of its legs to the aprons, two screws each, the same way as on the left.



1. Put it in place



2. Fasten it

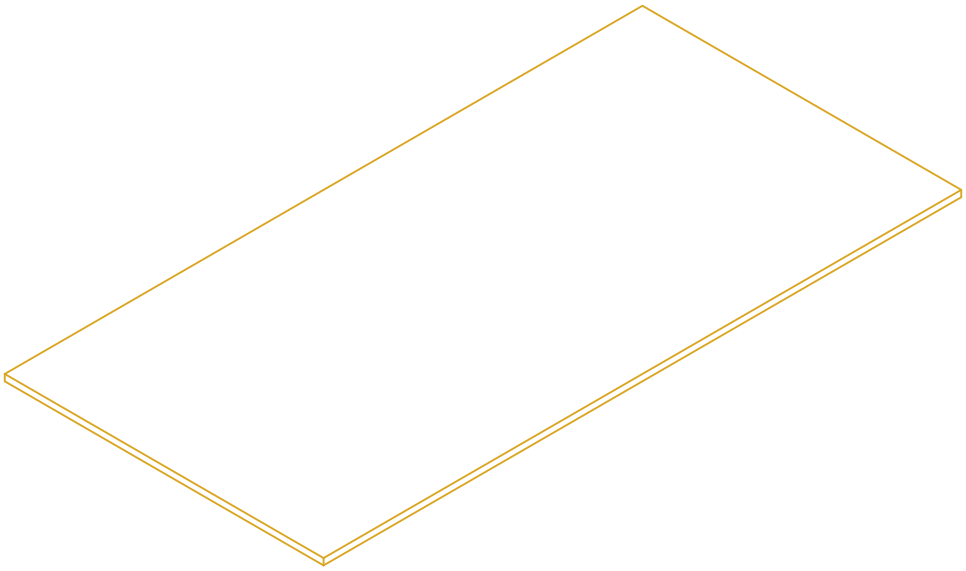
Joint: right-side rail-1 to back-apron end-b, pushed into place, up to 20 N.

Also: right-side rail-2 to front-apron end-a.

Fasteners: 4 × screws/sdws22600db into front-apron, back-apron through right-side, driven in, cutting its own thread.

MAKE

top



Top, 3/4 in. plywood cut to size

Package: //pub/furniture/workspace/basic
Needed: 1
Size: length 72, width 36
Made from: //pub/std/imperial/dimensional-lumber/stock:plywood/3-4x4x8

Made by taking material away from `//pub/std/imperial/dimensional-lumber/stock:plywood/3-4x4x8`.

Cut it to size with a saw:

1. Cut across X, 914.4 mm (36 in) from the -X end of the stock.

2. Cut across Y, 1828.8 mm (72 in) from the -Y end of the stock.

Step 4 of 4

Parts for this step

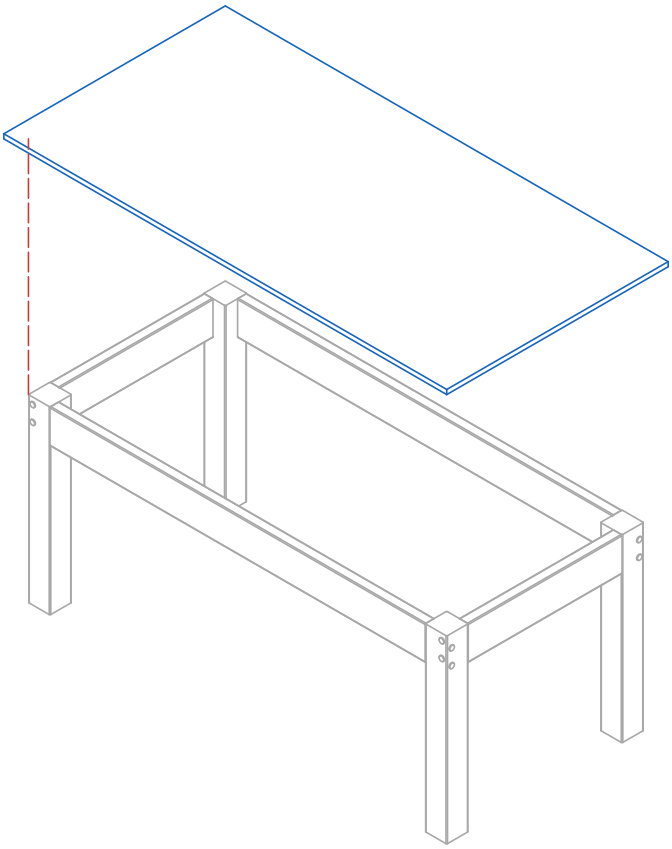


6 × screws/spax-8x2

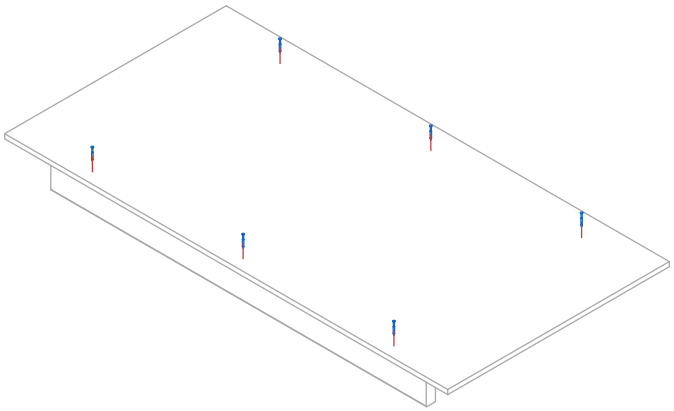
Lay the top on the frame, good face up, with 2 in. of overhang beyond the legs on every side.

It also meets right-side.

Drive three screws through the top into each long apron, 2-3/4 in. in from the edge of the top: 6 in. from each end of the apron and one halfway along. Sink the heads flush.



1. Put it in place



2. Fasten it

Joint: top front-left-leg to left-side leg-top, pushed into place, up to 20 N.

Also: top back-right-leg to right-side leg-top.

Fasteners: 6 × screws/spax-8x2 into front-apron, back-apron through top, driven in, cutting its own thread.

Links

This assembly: imperial-desk-1 *imperial-desk-1.assy*

This package: //pub/furniture/workspace/basic *README.md*

PartCAD *https://partcad.org/*